Determinant rule to calculate cross:

$$u = (u_x, u_y, u_z)$$

$$v = (v_x, v_y, v_z)$$

Doing:

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = (u_y v_z \hat{\imath} + u_z v_x \hat{\jmath} + u_x v_y \hat{k}) - (u_y v_x \hat{\imath} + u_x v_z \hat{\jmath} + u_z v_y \hat{k})$$

Alternatively, products go in the next form:

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} - \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = (u_x v_z \hat{\imath} + u_z v_x \hat{\jmath} + u_x v_y \hat{k}) - (u_z v_y \hat{\imath} + u_x v_z \hat{\jmath} + u_y v_x \hat{k})$$

In the example vectors:

$$u = (2, -1, 3)$$

$$v = (1, 4, -2)$$

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 3 \\ 1 & 4 & -2 \end{vmatrix} - \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 3 \\ 1 & 4 & -2 \end{vmatrix}$$

$$= ((-1)(-2)\hat{\imath} + 3 \cdot 1\hat{\jmath} + 2 \cdot 4\hat{k}) - (3 \cdot 4\hat{\imath} + 2 \cdot (-2)\hat{\jmath} + (-1) \cdot 1\hat{k})$$